Ish Kumar Jain

Assistant Professor
Electrical, Computer, and Systems Engineering
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Research Vision

My research focuses on advancing next-generation networks by optimizing spectrum resources to achieve high data rates, reliability, scalability, and practical deployments. I have particularly contributed in three areas:

- **Communication:** Develop innovative beamforming techniques (e.g., constructive multi-beams) to enhance reliable and scalable communication, particularly for mobile multi-user millimeter-wave networks.
- **Sensing:** Explore novel applications in wireless sensing, such as localization, tracking, and sensing-driven communication.
- **Security:** Investigate and mitigate security vulnerabilities (e.g., spoofing, jamming attacks) on wireless devices.

Professional Experience

Aug 2024 – present **Assistant Professor**, ***Rensselaer Polytechnic Institute (RPI)***, Albany, NY, USA.
Department of Electrical, Computer, and Systems Engineering
- Established a new wireless lab at RPI that focuses on "theory meets systems" research for building the next generation of wireless networks for communication, sensing, and security.

June–Dec 2022 **Research Intern**, ***VMware***, Palo Alto, CA, USA.
Mentor: Dr. Rakesh Misra
- Designed and patented a near-real-time application (xApps) with VMware RIC in Open-RAN framework.

June–Aug 2017 **Research Intern**, ***Nokia Bell Labs***, Murray Hill, NJ, USA.
Mentor: Dr. Özge Kaya
- Developed a Reinforcement Learning-based beam management scheme for mobile mmWave links, demonstrating 60% efficiency improvement over a baseline on a real-world mobility dataset.

Education

2018 - 2024 **University of California San Diego**, *CA*.
PhD, Electrical and Computer Engineering, *GPA: 4.0*
Advisor: Prof. Dinesh Bharadia

2016 - 2018 **New York University, Tandon School of Engineering**, *NY*.
MS, Electrical and Computer Engineering, *GPA: 3.96*
Advisor: Prof. Shiv Panwar

2012 - 2016 **Indian Institute of Technology (IIT Kanpur)**, *India*.
Bachelors of Technology (B.Tech.), Electrical Engineering, *GPA: 9.5 (out of 10)*

Awards and Honors

- NSF Award for IEEE Dyspan 2026 student travel grant: $15,000 during 2026.
- IBM-RPI Future of Computing Research Collaboration (FCRC) Award $125,000 during 2026.
- IBM-RPI Future of Computing Research Collaboration (FCRC) Award $150,000 during 2025.
- Best Thesis Presentation, ITA (Information Theory Workshop), San Diego, 2024.
- Best Poster Runner-up, Hotmobile 2023.
- VMware intern achievement award with $240,000 research grant 2022–23.
- Qualcomm Innovation Fellowship winner $100,000 research grant 2022–23.
- Marconi Society Scholar in Residence grant $5000 (sole recipient) 2021–22.
- Winner of 3-minute Thesis Award at ACM Mobisys'20, Mobicom'21, Mobicom'22.
- Commencement award for the best graduate student service in ECE, UC San Diego, May 2021.
- Commencement award for the best MS Academic Achievement in ECE, New York University, May 2019.
- Commencement award (Motorola Gold Medalist) for the best all-round performance in Electrical Engineering and Computer Science, IIT Kanpur, May 2016.
- Travel grant for MobiCom New Delhi 2018, Hotmobile CA 2023, Infocom NY 2023.

- Samuel Morse MS Fellowship ($100,000 financial support), New York University, 2016–2018.
- Secured All India Rank 390 (amongst 0.5 million students) in IIT–Joint Entrance Exam 2012.

Publications

TLDR: I published 15+ papers in top networking venues such as ACM Sigcomm, Usenix NSDI, ACM Sensys, and IEEE Journals. Multiple news media have covered our work and are supported by open-source code, posters, and demos.

ACM MobiSys 2026	BeamFormer: Transformer-based Beam Management for 6G Networks S Feng, S Kanjilal, K Qian, **IK Jain**
IEEE Infocom 2026	Satellites are closer than you think: A near field MIMO approach for Ground stations R R Vennam, L Wilson, **IK Jain**, D Bharadia
IEEE DySPAN 2026	Tiny-Twin: A CPU-Native Full-stack Digital Twin for NextG Cellular Networks A Mamaghani, U Ghosh, S Shakkottai, D Bharadia, **IK Jain**
ACM HotMobile 2026	SAPE: Demystifying Sub-band Aware Power-Equalization for Cellular Networks D Zhang, **IK Jain**, R Zhao
ACM Mobihoc 2025	FlexLink: Decoupling Control and Data Beams in Wideband Multi-antenna Networks **IK Jain**, R Vennam, D Bharadia – Media coverage
IEEE Infocom 2025	PhaseMO: Future-Proof, Energy-efficient, Adaptive Massive MIMO A Heidari, A Gupta, **IK Jain**, D Bharadia
SmallSat 2025	From Dishes to Distributed Arrays: Towards Efficient and Interference-Free Satellite Ground Stations *Small Satellite Conference (SmallSat) 2025, Salt Lake City, UT* R R Vennam, L Wilson, SSR Jonnavithula, **IK Jain**, D Bharadia
Mobicom Open-AI RAN 2024	MIMO-RIC: RAN Intelligent Controller for MIMO xApps *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking, Mobicom 2024* SSR Jonnavithula, **IK Jain**, D Bharadia
ACM Sensys 2024	CommRad: Collaborative Learning for Sensing-Driven mmWave Networks **IK Jain**, Suriyaa MM, D Bharadia (ACM Sensys 2024)
SmallSat 2024	mmSubArray: Enabling Joint Satellite and 5G Networks With Full-Spectrum Utilization in Millimeter-Wave Bands R Vennam, **IK Jain**, N Bhat, Suriyaa M, D Bharadia
ACM Hotmobile 2024	BeamArmor: Anti-Jamming in 5G Cellular Networks with MIMO Null-steering F Zumegen, **IK Jain**, D Bharadia
IEEE Infocom 2023	mmFlexible: Flexible Directional Frequency Multiplexing for Multi-user mmWave Networks **IK Jain**, RR Vennam, R Subbaraman, D Bharadia – Media coverage
IEEE S&P 2023	mmSpoof: Resilient Spoofing of Automotive Millimeter-wave Radars using Reflect Array *IEEE Security and Privacy 2023.* RR Vennam, **IK Jain**, K Bansal, J Orozco, P Shukla, A Ranganathan, D Bharadia – Media coverage
WPMC 2022	VRProj: Delivering 360-degree video with Viewport-adaptive Truncation *International Symposium on Wireless Personal Multimedia Communications (WPMC) 2022.* T Qiu, **IK Jain**, R Wu, P Cosman, D Bharadia
ACM HotCarbon 2022	Multiple smaller base stations are greener than a single powerful one: Densification of Wireless Cellular Networks, *ACM HotCarbon Workshop 2022* A Gupta, **IK Jain**, D Bharadia

ACM SIGCOMM 2021	[mmReliable] Two beams are better than one: Towards Reliable and High Throughput mmWave Links **IK Jain**, R Subbaraman, D Bharadia – Media coverage
ACM Mobicom mmNets 2020	mMobile: Building a mmWave Testbed to Evaluate and Address Mobility Effects *4th ACM Workshop on Millimeter-Wave Networks and Sensing Systems (Mobicom Workshop)*, 2020. **IK Jain**, R Subbaraman, TH Sadarahalli, X Shao, H Lin, D Bharadia
Usenix NSDI 2020	LocAP: Autonomous Millimeter Accurate Mapping of WiFi Infrastructure R Ayyalasomayajula, A Arun, C Wu, S Rajagopalan, S Ganesaraman, A Seetharaman, **IK Jain**, D Bharadia
MDPI Journal 2019	Extreme Multiclass Classification Criteria, *MDPI Computation Journal*, 2019. A Choromanska, **IK Jain**
IEEE JSAC 2018	The Impact of Mobile Blockers on Millimeter Wave Cellular Systems *IEEE Journal on selected areas in communications (JSAC)*, 2018 **IK Jain**, R Kumar, S Panwar
IEEE ITC 2018	Driven by Capacity or Blockage? A Millimeter-wave Blockage Analysis *IEEE International Teletraffic Congress (ITC) 2018* **IK Jain**, R Kumar, S Panwar – Invited paper

Posters/ Demos

Sensys 2025	[Demo] Millimeter-Wave Sub-Arrays for Joint Communications and Interference Suppression RR Vennam, L Wilson, **IK Jain**, D Bharadia
Mobicom 2024	[Demo] mm-O-RAN: Building a Millimeter-wave O-RAN Testbed S Mm, **IK Jain**, L Wilson, SSR Jonnavithula, D Bharadia
Mobicom 2024	[Demo] FlexLink Demo: Flexible Frequency-Dependent Multi-Beamforming with Delay-Phased Array **IK Jain**, YY Sun, S Cao, D Bharadia
Mobicom 2024	[Demo] BeamArmor5G: Demonstrating MIMO Anti-Jamming and Localization with srsRAN 5G Stack **IK Jain**, SSR Jonnavithula, D Bharadia
Mobicom 2024	[Poster] AppNet: Application-Aware Networking with O-RAN U Ghosh, **IK Jain**, S Seshasayee, S Shakkottai, D Bharadia
Hotmobile 2024	[Poster] Tiny-twin: A Lightweight and Verifiable Digital Twin for NextG Cellular Networks U Ghosh, **IK Jain**, S Shakkottai, D Bharadia
Hotmobile 2024	[Demo] Practical, Real-Time MIMO Apps with Programmable RAN Controller F Zumegen, **IK Jain**, D Bharadia
Milcom 2023	[Demo] BeamArmor: Anti-Jamming System in Cellular Networks with srsRAN Software Radios F Zumegen, **IK Jain**, D Bharadia
Mobicom 2023	[Demo] A Compact and Real-Time Millimeter-wave Experiment Framework with True Mobility Capabilities **IK Jain**, S MM, D Bharadia
Hotmobile 2023	[Poster] Delay Phased Arrays: Towards programmable beam-bandwidth for 5G networks **IK Jain**, RR Vennam, D Bharadia – Best Poster Runner-up
Mobicom S3 2021	[Demo and Dataset] for mmWave Multi-beam Tracking using mMobile 28 GHz Testbed **IK Jain**, R Subbaraman, D Bharadia
MobiCom 2018	[Poster] Facilitating Low Latency and Reliable VR over Heterogeneous Wireless Networks A Ravichandran, **IK Jain**, R Hegazy, T Wei, D Bharadia

Patents

US Patent (provisional) — Collaborative Learning for Sensing-Driven Millimeter-wave Networks
D Bharadia, **IK Jain**, et al.

US Patent (provisional) — Method and Apparatus for Resilient Spoofing of Automotive Radars
D Bharadia, **IK Jain**, et al.

US Patent (granted) — Flexible directional frequency multiplexing for multi-user RF networks
IK Jain, RR Vennam, R Subbaraman, D Bharadia - US Patent 19/109,686, 2026

US Patent (granted) — Ran applications for inter-cell interference mitigation for massive MIMO in a RAN
Y Yang, N Paranjape, D Muthunoori, **IK Jain** - US Patent 12,457,594, 2025

US Patent (granted) — Enabling Reliable Millimeter-Wave Links using Multi-Beam, Pro-active Tracking, and Phased Array Designs
IK Jain, D Bharadia, T Sadarahalli, R Subbaraman - US Patent 12,519,517, 2026

US Patent (granted) — Enable Indoor Navigation with Context assisted Localization
R Ayyalasomayajula, D Bharadia, S Ganesaraman, **IK Jain**, S Rajagopalan, A Seetharaman, S Sharma, A Arun, C Wu - US Patent 12,282,111, 2025

Teaching and Mentoring Experience

Teaching .

- Fall 2025-26 — ECSE 4560/6560- Modern Communication System (Graduate+Undergraduate)
- Spr 2025-26 — ECSE 4660/6660- Internetworking of Things (Graduate+Undergraduate)
- Fall 2024 — ECSE 4960/6960- Seminar on AI for Networking (Graduate+Undergraduate)

Teaching Assistant.

- WI 2021,23 — ECE 257B- Modern Wireless Communication (Graduate) – Best TA 10/10 rating [Link to class evaluations]
- Spring 2020 — ECE 157B- Wireless Sensing Systems (Undergraduate) – Helped design a new class [Link to my notes]
- Spring 2018 — EEGY 9123- Introduction to Machine Learning (Graduate)
- Fall 2017 — EEUY 4563- Introduction to Machine Learning (Undergraduate) – Helped design a new class[Link to class exercises]
- Spring 2017 — ELGY 6373- Internet Architecture and Protocols Lab (Graduate)

Mentorship Experience, I have mentored historically underrepresented and underprivileged students.

- 2026-Ongoing — Andrew Nguyen (RPI PhD)
- 2025-Ongoing — Swastik Kanjilal (RPI PhD)
- 2024-Ongoing — Raj Bakhunchhe (RPI PhD, co-advised)
- 2022-24 — Rohith Reddy Vennam (UCSD PhD)
- 2023-24 — Ushasi Ghosh (UCSD PhD)
- 2023-24 — Mohamed Waeel (UCSD PhD)
- 2022-24 — Suriyaa MM (UCSD MS → Apple)
- 2021-23 — Joshua Orozco (UCSD MS → TrellisWare Technologies)
- 2021-23 — Puja Shukla (UCSD MS → Marvel Technology)
- 2019-22 — Tian Qiu (UCSD MS → UCSB PhD)
- 2020-22 — Xiangwei Shao (UCSD MS → Huawei, China)
- 2020-22 — Weginbara (Michael) Youpele (UCSD MS → Naval Surface Warfare Center)
- 2019-21 — Hou-Wei Lin (UCSD MS → Amazon)
- 2018-20 — Tejas Sadarahalli (UCSD MS → Qualcomm)
- 2019-21 — Raini Wu (UCSD BS → UCSD PhD)

Invited Talks

University and Industry Talks.

Sep 2025 — Rethinking Next-G Networks for Communication and Sensing
Osaka University, Osaka, Japan (invited talk, in-person)

Sep 2025 Rethinking Next-G Millimeter-wave Networks for Communication, Sensing, and Security
University of Georgia (UGA), GA (virtual)

Aug 2025 Rethinking Next-G Millimeter-wave Networks for Communication, Sensing, and Security
Samsung Research America, Dallas, TX (invited talk, in-person)

Mar 2025 Rethinking Next-G Millimeter-wave Networks for Communication, Sensing, and Security
Texas A&M University, College Station, TX

Nov 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
BITS Pilani, Pilani Campus, India

Nov 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
IIT Roorkee, India

Nov 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
IIIT Hyderabad, India

Nov 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
IIT Hyderabad, India

Nov 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
IIT Madras, India

Sep 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
University at Buffalo, SUNY, Buffalo, NY

Sep 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
Northeastern University, Boston, MA

Mar 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
University of Texas, Austin, TX

Mar 2024 Reliable, Scalable, and Secure 6G Networks for Communication and Sensing
Rensselaer Polytechnic Institute, Troy, NY

Nov 2023 Reliable, Scalable and Deployable Millimeter-wave System
IMDEA Network Institute, Madrid, Spain

Aug 2023 Reliable, Scalable and Deployable Millimeter-wave System
Qualcomm Innovation Fellowship talk, San Diego, CA

Jul 2023 Rethinking Next-G Millimeter-wave Networks
Indian Institute of Science (IISc), Bangalore, India

Jun 2023 Rethinking Next-G Millimeter-wave Networks
Indian Institute of Technology (IIT) Kanpur, India

Jun 2023 Rethinking Next-G Millimeter-wave Networks
Carnegie Mellon University, Pittsburgh, PA

May 2023 Rethinking Next-G Millimeter-wave Networks
Princeton University, NJ

May 2023 Rethinking Next-G Millimeter-wave Networks
New York University, NY

Apr 2023 Towards reliable mmWave links
University of Washington, Seattle, WA

Sep 2022 Building xApps with VMWare RIC
VMware, Palo Alto, CA (invited talk)

Jun 2022 Towards reliable and high-throughput mmWave links
University of Colorado, Denver, CO

Conference Talks.

Oct 2025 FlexLink: Decoupling Control and Data Beams in Wideband Multi-antenna Networks
Mobihoc 2025, Houston, TX

Nov 2024 CommRad: Collaborative Learning for Sensing-Driven mmWave Networks
Sensys 2024, Hangzhou, China (virtual)

Oct 2024 BeamArmor5G: Demonstrating MIMO Anti-Jamming and Localization with srsRAN 5G Stack
srsRAN Workshop, Washington DC

Mar 2024 BeamArmor: Anti-Jamming in 5G Cellular Networks with MIMO Null-steering
Hotmobile 2024, San Diego, CA

Mar 2024 Reliable, Scalable and Deployable Millimeter-wave System
ITA–Information Theory Workshop 2024, San Diego, CA (Best Presentation Award)

Nov 2023 A Compact and Real-Time Millimeter-wave Experiment Framework with True Mobility Capabilities
Mobicom mmNets 2023, Madrid, Spain

May 2023 mmFlexible: Flexible Directional Frequency Multiplexing for Multi-user mmWave Networks
Infocom 2023, New York City, NY

Aug 2021 Reliable Multi-user Millimeter-wave Networks
Mobicom Workshop 2022 (virtual, Best 3-minute Thesis Award)

Aug 2021 Reliable mmWave Networks
Mobicom Workshop 2021, New Orleans, LA (Best 3-minute Thesis Award)

Aug 2021 Two beams are better than one: Towards Reliable and High Throughput mmWave Links
Sigcomm 2021 (virtual)

Aug 2021 Towards reliable mmWave networks
Mobisys Workshop 2020 (virtual, Best 3-minute Thesis Award)

Aug 2020 mMobile: Building a mmWave Testbed to Evaluate and Address Mobility Effects
Mobicom mmNets 2020 (virtual)

Leadership

2021–2022 **The Marconi Society**, *Scholar in Residence*.
Served as a student scholar for facilitating the Marconi Society meetings with the chair Vint Cerf and other prominent scientists, engineers, and policymakers.

2021–2022 **Escribamos Ciencia K12 team**, *UC San Diego*.
Developed interesting science modules and videos for K12 students on topics such as electricity, internet, nuclear power, etc., under the guidance of Prof. Olivia Graeve.

2020–2022 **Jacobs Undergraduate Mentorship Program**, *UC San Diego*.
Mentored underprivileged students through lab tours, industry talks, panel discussions, technical workshops, etc., and bridged the communication gap between undergraduates and graduate students.

2020–2021 **O.W.L Reading group**, *Inter-continental collaboration*.
Founded in Fall 2020 as a small group of Ph.D. students interested in research talks and discussion on recent conference papers and has grown to over 100+ members in a year.

2019–2021 **Vice President, ECE graduate student council**, *UC San Diego*.
Responsible for providing communication between ECE students and the ECE department and organizing seminars and student mentorship programs.

Services

2023-2025 **Co-chair**.
ACM Mobicom WinTech workshop 2026, Co-chair
ACM Mobicom 2026, Registration Chair
IEEE DySpan 2026, Student Travel Grant chair
ACM Mobihoc 2025, Registration Chair
IEEE International Conference on Network Protocols (ICNP 2025), Publicity chair
ACM Mobicom Open-AI-RAN Workshop 2024, 2025, Publicity chair
ACM Mobicom S3 Workshop 2023, TPC co-chair
ACM Mobicom mmNets 2023, Publicity chair

2021–2025 **Technical Program Committee (TPC)**.
IEEE ICNP 2026
ACM Mobicom 2025, 2026
ACM Mobisys 2025, 2026
IEEE ICASSP 2025
IEEE WCNC 2025
IEEE MSN 2024
IEEE WCNC 2022
ACM Mobicom S3 Workshop 2021

2019–2024 **Artifact Evaluation Committee (AEC)**.
ACM Mobicom 2023, 24
ACM Sigcomm 2023
ACM CoNEXT 2019

2019–2025 **Technical Reviews**.
IEEE Transactions on Signal Processing (TSP) 2025
IEEE Communications Magazine 2023, 25
IEEE ISIT 2024
IEEE Transactions on Mobile Computing (TMC) 2024
IEEE Transactions on Communications (TCOM) 2023
IEEE Communications Letters 2022, 23
IEEE Transactions Vehicular Technology (TVT) 2019, 20, 24
ACM/ IEEE Transactions on Networking (TNET) 2021, 23, 24, 25
IEEE Access 2021- 23
IEEE Journal on Selected Areas in Communication (JSAC) 2023
IEEE WCNC 2022-23

2021–2022 **Lead organizer and moderator**.
Sigcomm'21 Social Trivia Organizer
COMSNETS'22 Panel Moderator